

ECOTEC_Online Student Risk Dataset

Codebook and Metadata Documentation

This codebook describes the structure, variables, metadata, target provenance, and anonymization procedures associated with the ECOTEC_Online Student Risk Dataset. The repository supports reproducible research in learning analytics, educational data mining, institutional analytics, and responsible artificial intelligence.

Dataset Overview

Attribute	Description
Institution	Universidad ECOTEC
Country	Ecuador
Educational modality	Online higher education
Number of records	12,632
Dataset type	Institutional tabular dataset
Target variable	TARGET_RISK

Variable Dictionary

Variable	Description	Type	Missing	Unique values
PUBLIC_RECORD_ID	Release-specific anonymous record identifier	Numeric	0	12,632
FACULTY	Academic faculty	Categorical/Text	0	1
DEGREE_PROGRAM	Academic degree program	Categorical/Text	0	29
CURRICULUM_YEAR	Curriculum year	Numeric	0	4
GENDER	Student gender	Categorical/Text	0	2
AGE	Student age	Numeric	0	62
MARITAL_STATUS	Student marital status	Categorical/Text	0	6
ETHNICITY	Self-reported ethnicity	Categorical/Text	0	6
DISABILITY	Disability condition indicator	Categorical/Text	0	2
INSTITUTION_TYPE	Type of prior or associated institution	Categorical/Text	0	5
NATIONALITY_COUNTRY	Nationality country category	Categorical/Text	0	9
TRANSFER_STUDENT_FLAG	Transfer-student indicator	Categorical/Text	0	2
PAYMENT_PLAN	Institutional payment-plan category	Categorical/Text	0	2
ONLINE_ADMISSION_TYPE	Online admission pathway	Categorical/Text	0	15
PAYMENT_DEFERRAL_TYPE	Institutional payment-deferral category	Numeric	0	25
SCHOLARSHIP_NAME	Scholarship or discount category	Categorical/Text	0	9
TARGET_RISK	Binary administrative target derived during institutional curation. Class 0 indicates that alumni status or a valid graduation date was recorded; class 1 indicates that the student was not recorded as alumni and no graduation date was available. The variable is contemporaneous and does not constitute a prospective dropout or retention outcome.	Numeric	0	2
REGION	Geographic region category	Categorical/Text	0	4

TARGET_RISK Operational Definition and Provenance

TARGET_RISK is a binary administrative target derived during institutional curation from two restricted institutional source variables: STUDENT_TYPE and GRADUATION_DATE.

Class	Operational rule	Interpretation at extraction
0	Alumni status or a valid graduation date was recorded.	Completion status documented.
1	The student was not recorded as alumni and no graduation date was available.	Completion status not documented.

Temporal interpretation. The label and the released predictors correspond to the same institutional snapshot. The target is contemporaneous and does not constitute a prospective dropout or retention outcome.

Predictor separation. STUDENT_TYPE and GRADUATION_DATE were excluded from the public dataset and from the benchmark predictor matrix after target construction. The released predictors therefore do not directly contain the two source fields used to define the target.

Data Anonymization

Personally identifiable information was removed prior to public dissemination. The anonymization workflow excluded names, identification numbers, email addresses, telephone numbers, and direct institutional identifiers. Additional transformations and categorical aggregations were applied to reduce re-identification risk while preserving analytical utility.

FAIR Data Principles

The repository was organized following FAIR data principles to facilitate findability, accessibility, interoperability, and reuse of educational AI resources.

Citation

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